

FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

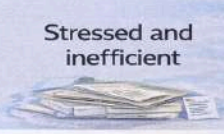
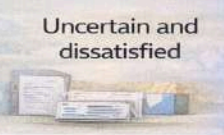
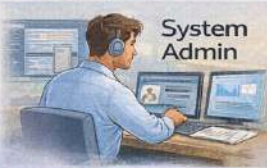
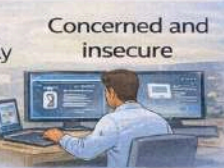
- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

Problem Statements – Farm Management System

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	
PS-2	 Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	 Stressed and inefficient
PS-3		Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	 Uncertain and dissatisfied
PS-4	 System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	 Concerned and insecure

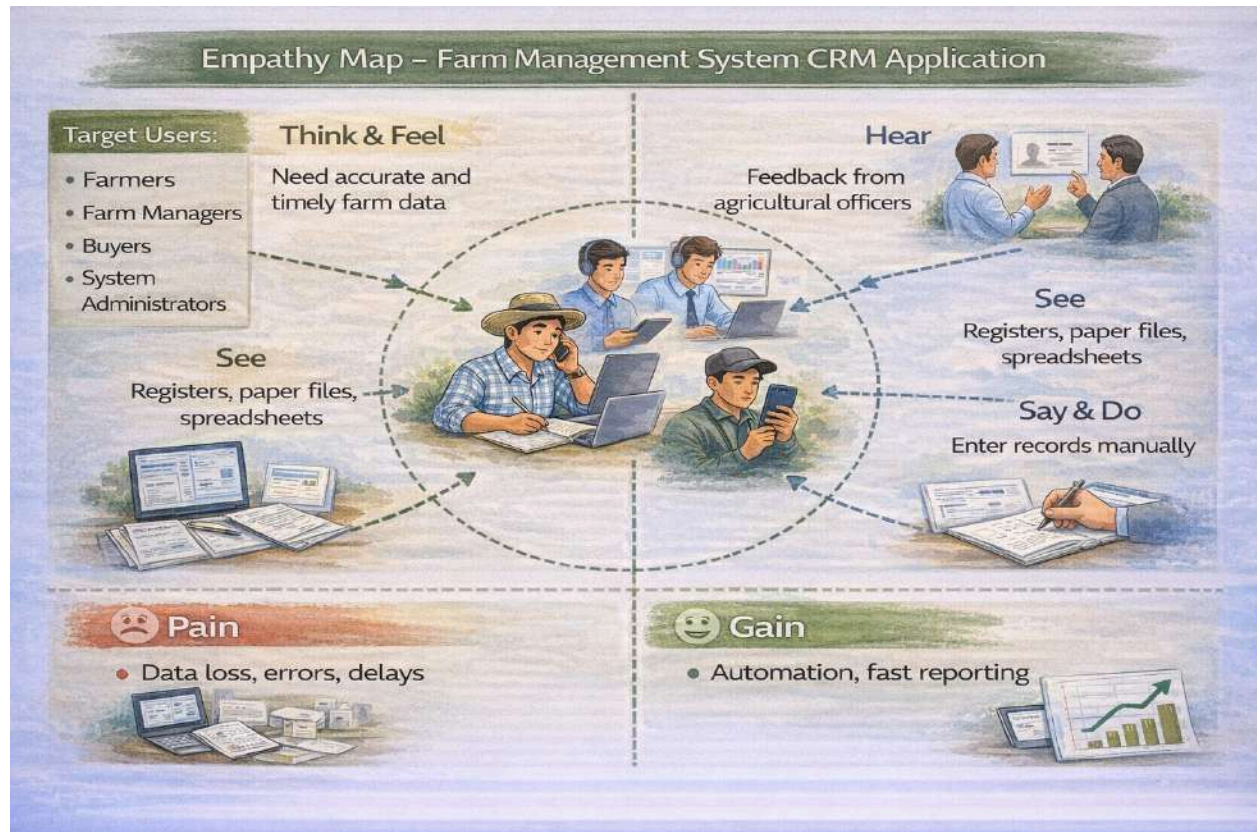
2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



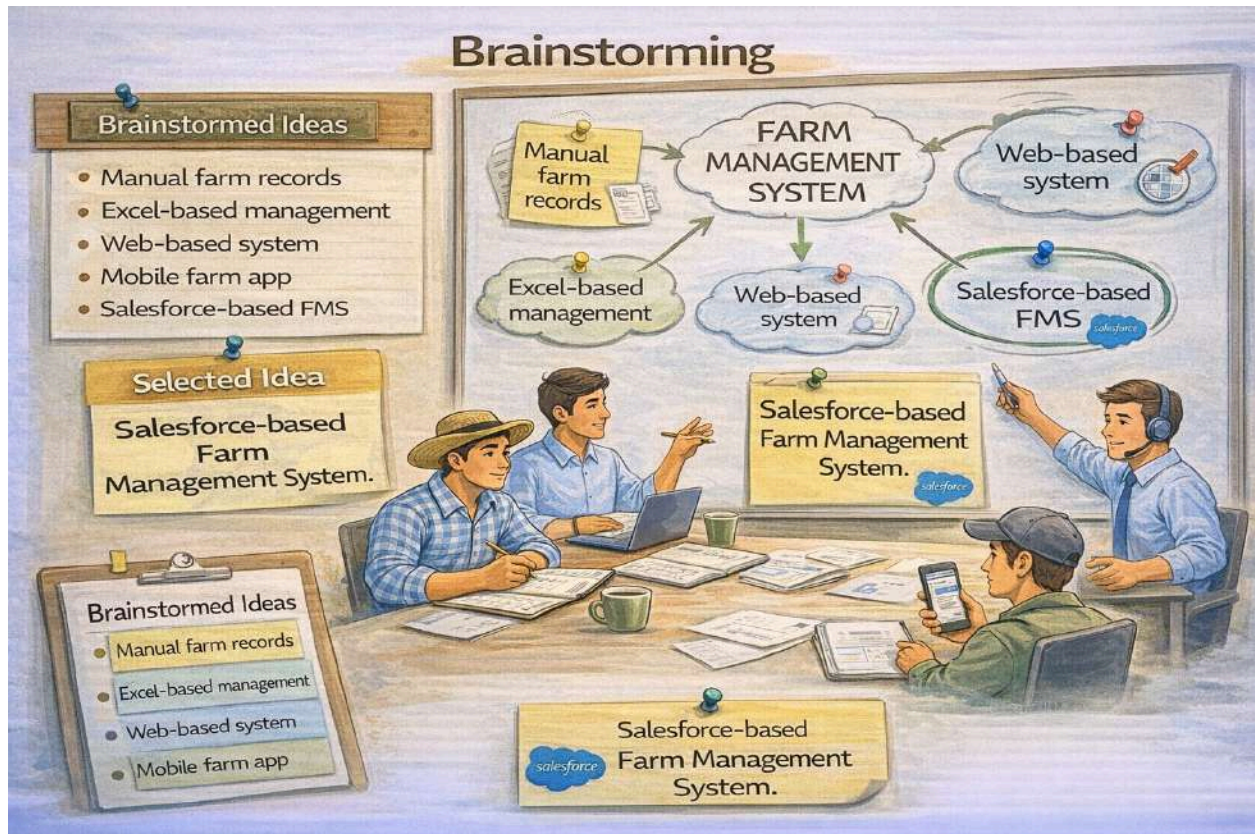
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

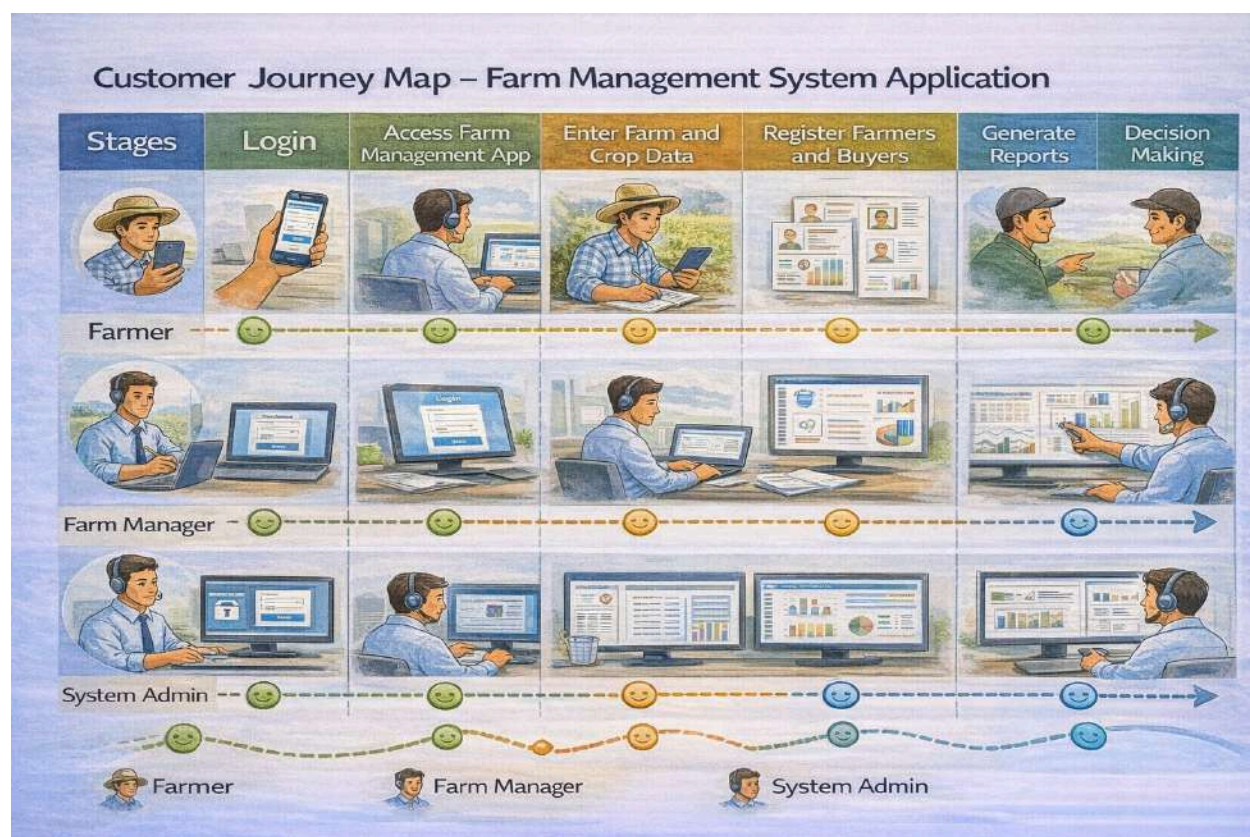


3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation

FR-6	Dashboard Visualization
FR-7	User Role Management

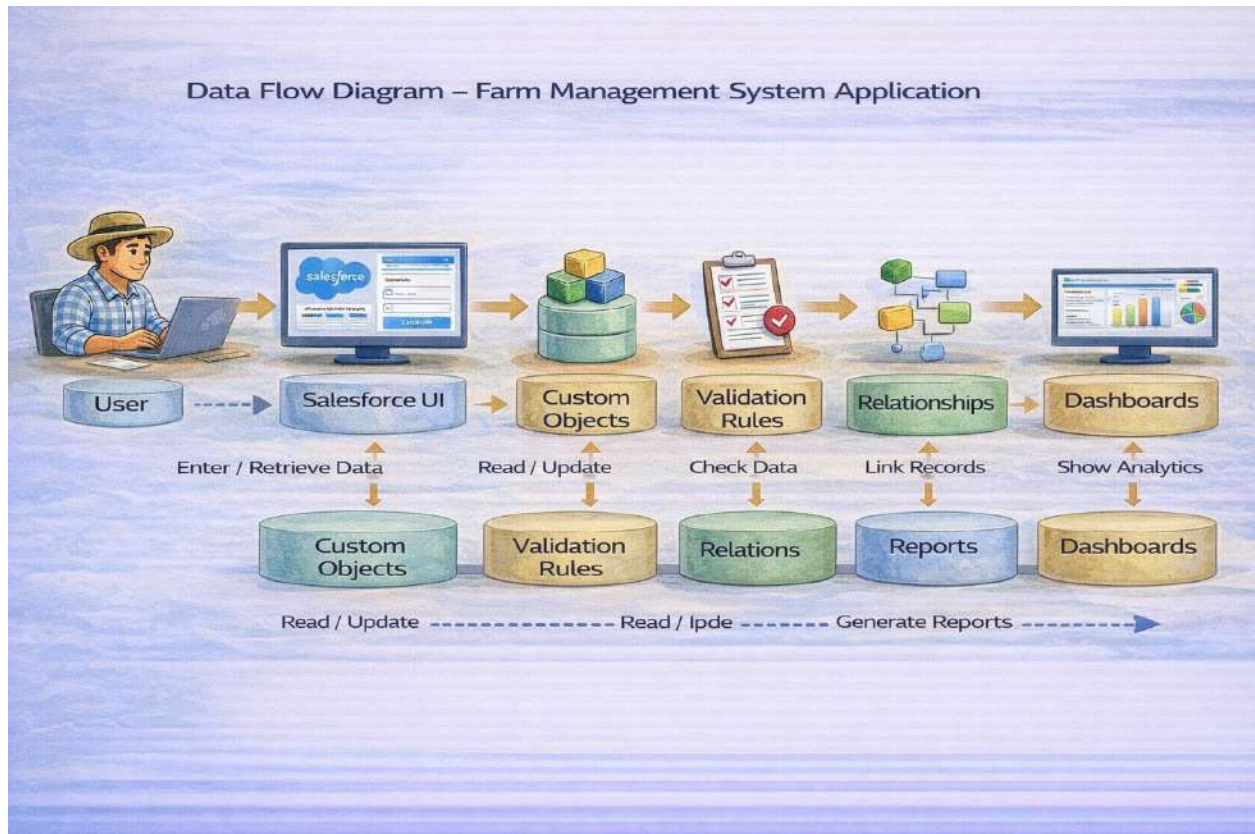
Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



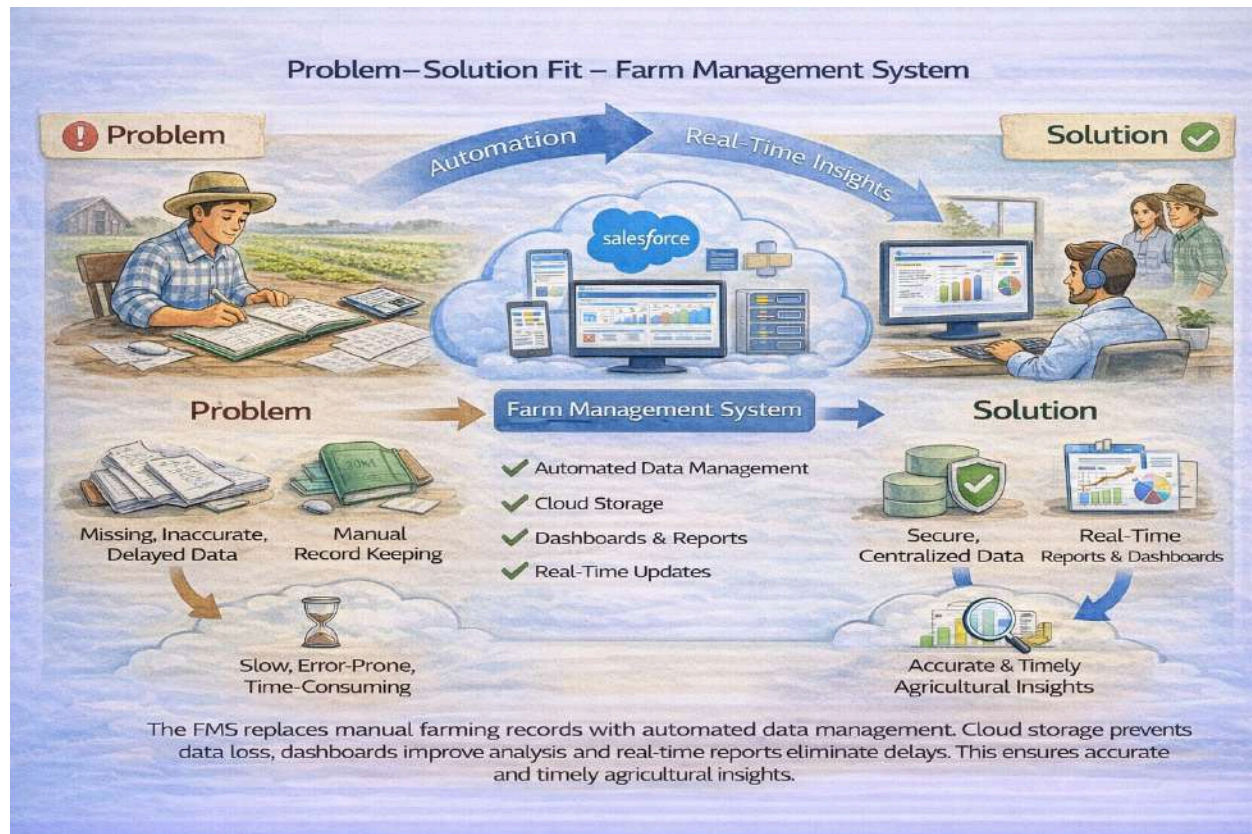
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



4.2 Proposed Solution

Functional Requirements

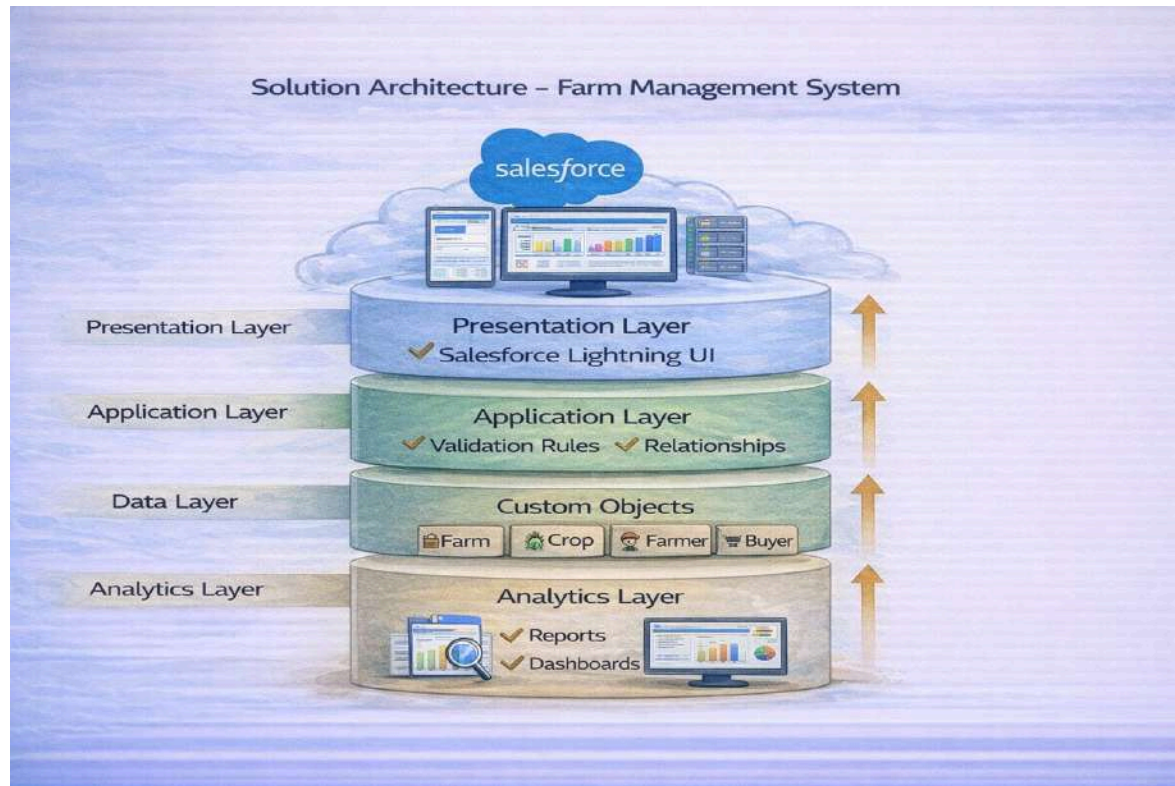
Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.

Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.es
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will receive a mail from salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com

Build enterprise-quality apps fast to bring your ideas to life

- Build apps fast with drag and drop tools
- Customize your data model with clicks
- Go further with Apex code
- Integrate with anything using powerful APIs
- Stay protected with enterprise-grade security
- Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*
Your first name

Last Name*
Your last name

Email*
Your email address

Role*
Your job title

Company*
Company Name

Country/Region*
Country/Region

Postal Code*
Your postal code

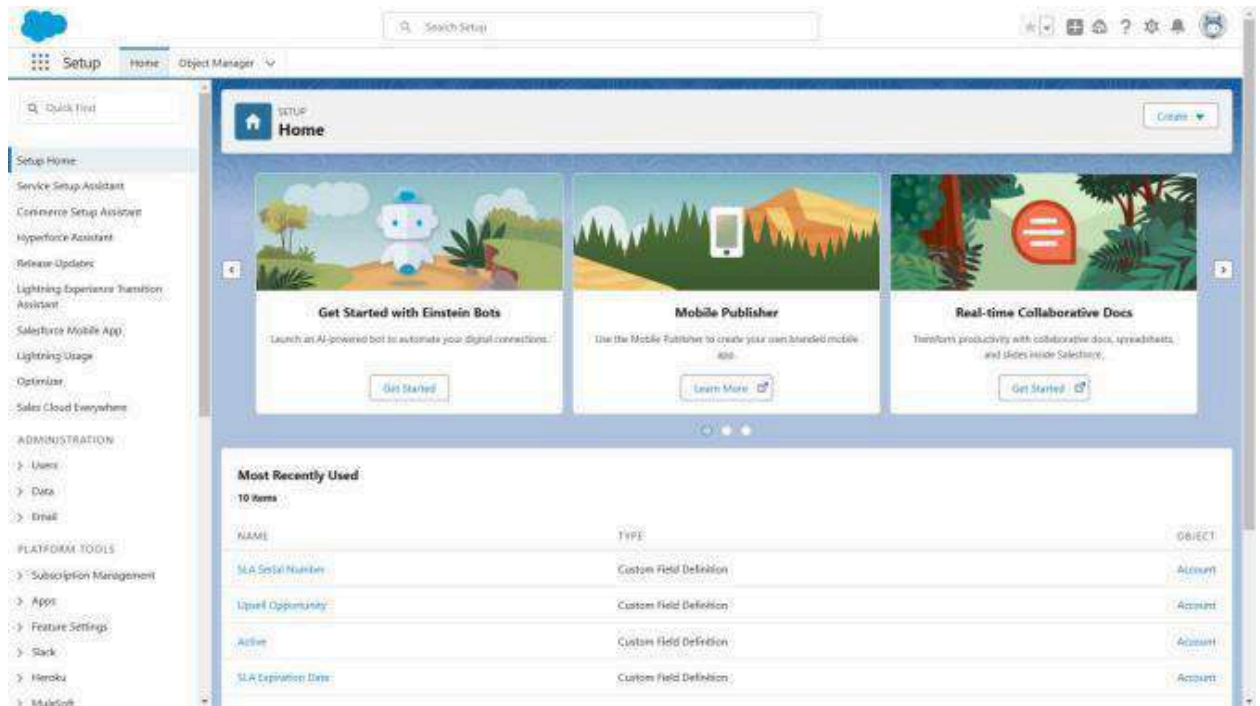
Username*
jane@company.sandbox

Your username must be in the form of an email address. It does not have to be real. It must be unique and cannot be associated with another Salesforce login instance. Read more about username requirements.

☐ I agree to the [Master Services Agreement](#) – Developer Services and [Salesforce Program Agreement](#).

Sign me Up

6. By using username and password you can into the Salesforce org. The homepage will appear below



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".

The screenshot shows the Salesforce Object Setup page for an object named 'Farm'. The left sidebar contains a navigation menu with the following items: Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Recordion Rules, Scoping Rules, and Object Access. The main content area is titled 'Data Type' and shows 'Text' selected. A warning message states: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.' The page is divided into several sections: 'Optional Features' with checkboxes for 'Allow Reports' (checked), 'Allow Activities' (unchecked), 'Track Field History' (unchecked), 'Allow in Chatter Groups' (unchecked), and 'Enable Licensing' (unchecked); 'Object Classification' with checkboxes for 'Allow Tracking' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked); 'Deployment Status' with radio buttons for 'In Development' (selected) and 'Deployed' (unchecked); and 'Search Status' with a checkbox for 'Allow Search' (checked). At the bottom right, there is a 'What's New?' link. At the bottom center, there are three buttons: 'Save', 'Save & New', and 'Cancel'.

7. In the Optional Features section, select Allow Reports and Track Field History.
 8. In the Deployment Status section, ensure Deployed is selected.
 9. In the Search Status section, select Allow Search.
 10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
-
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox
 - g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
 - i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer

- c. Plural Label: Farmers
- d. Record Name: Farmer Name e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Milestone 3 -

Tab: Tab

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

Custom Tabs

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Action	Label	Tab Style	Description
New			
New	Accounts	Standard	
New	Contacts	Standard	
New	Leads	Standard	
New	Opportunities	Standard	
New	Products	Standard	
New	Users	Standard	
New	Roles	Standard	
New	Profiles	Standard	
New	Groups	Standard	
New	Jobs	Standard	
New	Students	Standard	

2. Under custom object tabs, click New.

Cloud logo | Search Setup | Setup | Home | Object Manager

Setup | Tabs

New Custom Object Tab

Help on this Page

Step 1 of 3: Enter the Details

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab [Get Required Information](#)

Select an existing custom object or create a new custom object now.

Object:

Tab Style:

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Custom Page Custom Link:

Enter a short description

Description:

3. For Object, select Farm.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

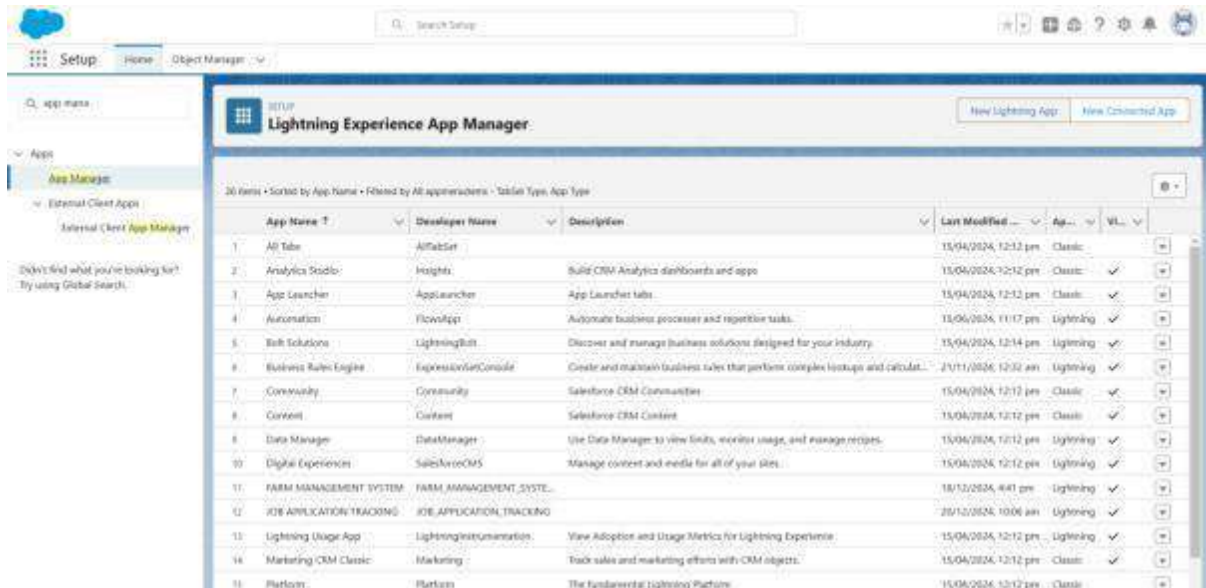
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app



The screenshot shows the Salesforce Lightning Experience App Manager interface. The left sidebar contains navigation links for Setup, Home, and Object Manager. The main content area displays a list of 20 apps, sorted by App Name. The table includes columns for App Name, Developer Name, Description, Last Modified, App Type, and Visibility. The 'FARM MANAGEMENT SYSTEM' app is highlighted in the list.

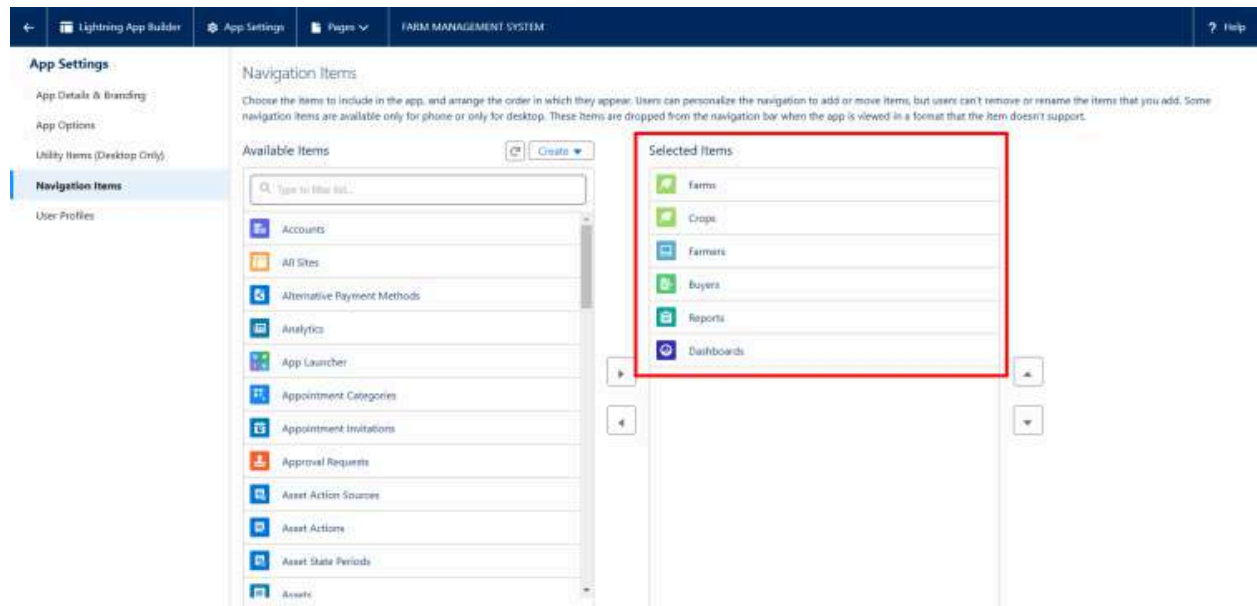
App Name	Developer Name	Description	Last Modified	App Type	Visibility
1. All Tabs	AllTabs		15/04/2024, 12:12 pm	Classic	
2. Analytics Studio	Insights	Build CRM Analytics dashboards and apps	15/04/2024, 12:12 pm	Classic	✓
3. App Launcher	AppLauncher	App Launcher tabs	15/04/2024, 12:12 pm	Classic	✓
4. Automation	FlowApp	Automate business processes and repetitive tasks	15/04/2024, 11:17 pm	Lightning	✓
5. Built Solutions	LightningBolt	Discover and manage business solutions designed for your industry	15/04/2024, 12:14 pm	Lightning	✓
6. Business Rules Engine	ExpressionEngine	Create and maintain business rules that perform complex lookups and calculations	21/11/2024, 12:02 am	Lightning	✓
7. Community	Community	Salesforce CRM Communities	15/04/2024, 12:12 pm	Classic	✓
8. Content	Content	Salesforce CRM Content	15/04/2024, 12:12 pm	Classic	✓
9. Data Manager	DataManager	Use Data Manager to view facts, monitor usage, and manage recipes	15/04/2024, 12:12 pm	Lightning	✓
10. Digital Experience	SalesforceCMS	Manage content and media for all of your sites	15/04/2024, 12:12 pm	Lightning	✓
11. FARM MANAGEMENT SYSTEM	FARM MANAGEMENT SYSTEM		18/12/2024, 4:41 pm	Lightning	✓
12. JDE APPLICATION TRACKING	JDE APPLICATION TRACKING		20/12/2024, 10:06 am	Lightning	✓
13. Lightning Usage App	LightningInstrumentation	View Adoption and Usage Metrics for Lightning Experience	15/04/2024, 12:12 pm	Lightning	✓
14. Marketing CRM Classic	Marketing	Track sales and marketing efforts with CRM objects	15/04/2024, 12:12 pm	Classic	✓
15. Platform	Platform	The fundamental Lightning Platform	15/04/2024, 12:12 pm	Classic	

- From Setup, enter App Manager in the Quick Find and select App Manager.
 - Click New Lightning App. Farm Management System as the App Name, then click Next
 - Under App Options, leave the default selections and click Next.
 - Under Utility Items, leave as is and click Next.
 - From Available Items, select Farm, Crop,Farmer,Buyer Reports,and Dashboards and move them to Selected Items. Click Next. ● From Available Profiles, select System Administrator and move it to Selected Profiles.Click Save & Finish.
 - To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.
- Note:
1. App Launcher-Displays available apps.
 2. App Name-Displays the current selected app.

The screenshot shows the 'New Lightning App' configuration interface. The 'App Details' section on the left includes a red-bordered input field for 'App Name' with the placeholder 'Name your app...'. Below this are fields for 'Developer Name' and 'Description'. The 'App Branding' section on the right features an 'Image' upload button, a 'Primary Color Hex' dropdown set to '#0070C0', and a 'Use Theme Options' checkbox. A blue 'Next' button at the bottom right is also highlighted with a red box. The background displays a table of existing apps, including 'Lightning Image App', 'Marketing CRM Classic', and 'Platform'.

App Name	Category	App Description	Created By	Created Date	Status
Lightning Image App	Lightning Experience	View, update and create content for Lightning Experience	System Administrator	10/12/2016	Active
Marketing CRM Classic	Marketing	Track sales and marketing efforts with CRM records	System Administrator	11/04/2016	Classic
Platform	Platform	The fundamental Lightning Platform	System Administrator	11/04/2016	Classic

3. Navigation menu-Displays the tabs available inside the app.



Milestone 5 -

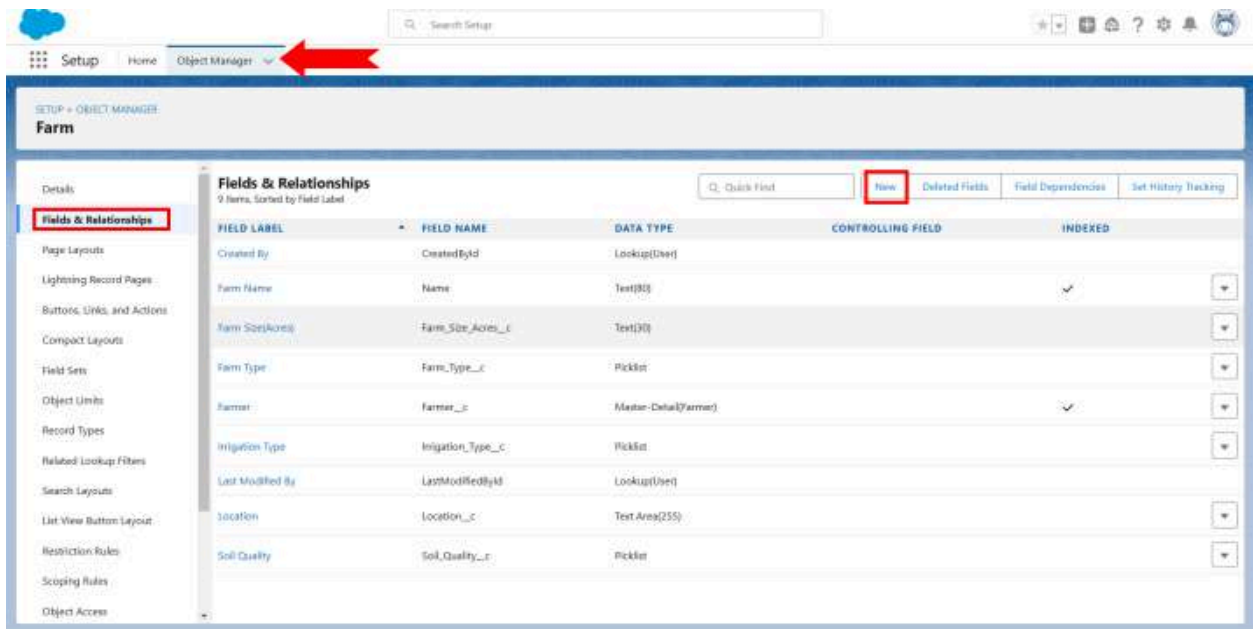
Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



The screenshot shows the Salesforce Setup page for the Farm object. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. A red arrow points to the 'Object Manager' tab. The left sidebar shows the 'Fields & Relationships' section highlighted. The main content area displays a table of fields for the Farm object, with columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. A 'New' button is visible in the top right of the Fields & Relationships section.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Farm Name	Name	Text(80)		✓
Farm Size (Acres)	Farm_Size_Acres__c	Text(30)		
Farm Type	Farm_Type__c	Picklist		
Farmer	Farmer__c	Master-Detail(Farmer)		✓
Irrigation Type	Irrigation_Type__c	Picklist		
Last Modified By	LastModifiedBy	Lookup(User)		
Location	Location__c	Text Area(255)		
Soil Quality	Soil_Quality__c	Picklist		

5. Choose Data type as a picklist, Click next

The screenshot shows the Salesforce Setup interface for the 'Farm' object. The left sidebar contains a navigation menu with the following items: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'Farm' and displays a list of data types with their descriptions. The 'Picklist' option is selected and highlighted with a red box. The descriptions for the data types are as follows:

- Checkbox**: Allows users to select or deselect a value (primarily or a non-primarily value).
- Currency**: Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
- Date**: Allows users to enter a date or pick a date from a popup calendar.
- DateTime**: Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the DateTime field.
- Email**: Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.
- Geolocation**: Allows users to define locations, includes latitude and longitude components, and can be used to calculate distance.
- Number**: Allows users to enter any number. Leading zeros are removed.
- Percent**: Allows users to enter a percentage number, for example, "10" and automatically adds the percent sign to the number.
- Phone**: Allows users to enter any phone number. Automatically formats it as a phone number.
- Picklist**: Allows users to select a value from a list you define.
- Picklist (Multi-Select)**: Allows users to select multiple values from a list you define.
- Text**: Allows users to enter any combination of letters and numbers.
- Text Area**: Allows users to enter up to 255 characters on separate lines.
- Text Area (Long)**: Allows users to enter up to 131,072 characters on separate lines.
- Text Area (Rich)**: Allows users to enter formatted text, add images and links. Up to 131,072 characters on separate lines.
- Text (Encrypted)**: Allows users to enter any combination of letters and numbers and store them in encrypted form.
- Time**: Allows users to enter a local time. For example, "2:40 PM", "14:40", "14:40:00", and "14:40:00.000" are all valid times for this field.
- URL**: Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

At the bottom right of the page, the 'Next' button is highlighted with a red box, and the 'Cancel' button is visible next to it.

6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)
7. Next, Next Click Save.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a navigation menu with the following items: Details, Fields & Relationships (highlighted), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'New Custom Field' and is at 'Step 2 of 4: Enter the details'. A red rectangle highlights the 'Field Label' and 'Values' sections. The 'Field Label' is 'Farm Type'. The 'Values' section has the option 'Use global picklist value set' unchecked and 'Enter values, with each value separated by a new line' selected. The text area for values contains 'Crop', 'Livestock', and 'mixed'. Below this, the options 'Display values alphabetically, not in the order entered' and 'Use first value as default value' are unchecked, while 'Restrict picklist to the values defined in the value set' is checked. At the bottom, the 'Field Name' is 'Farm_Type', and the 'Description' and 'Help Text' fields are empty.

Setup Home Object Manager

Search Setup

SETUP > OBJECT MANAGER

Farm

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

New Custom Field

Step 2 of 4: Enter the details

Field Label: Farm Type

Values:

- ☐ Use global picklist value set
- ☒ Enter values, with each value separated by a new line

Crop
Livestock
mixed

☐ Display values alphabetically, not in the order entered

☐ Use first value as default value

☒ Restrict picklist to the values defined in the value set

Field Name: Farm_Type

Description:

Help Text:

Previous Next Cancel

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

- 1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
- 2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
- 3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
- 5.Create a Master Detail relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

- 1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
- 3.Again we select data type as a date,Enter field labe (Harvest date),save
- 4.Next We Create Lookup relationship on crop object realted to Farm Object.
- 5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.
- 6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
- 7.Now We create Formula Field For That Select data Type As a Formula,Click Next.

SETUP & OBJECT MANAGER

Crop

Charpelle

Fields & Relationships

Page Layout

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Data

Object Units

Harvested Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scrimping Rates

Object Address

Specify the type of information that the custom field will contain

Data Type

☐ None Selected

Select one of the data types below:

 Auto Number

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

439/5000

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

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A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records linked in a related list.

Looking Relatively

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a jump list. The other object is the source of the values in the list.

 [More Detail Information](#)

Creates a special type of parent-child relationship between the object (the child, or "derail") and another object (the parent, or "master") where the child is not a true child of the parent.

- The relationship field is required in all detail records.
 - The ownership and drawing of a detail record are determined by the master record.
 - When a user deletes the master record, all detail records are deleted.
 - You can create rolling summary fields on the master record to summarize the detail records.
- The relationship field allows users to click on a toolbar icon to select a value from a census by

The relationship field allows users to click on a lookup icon to select a value from a popup list.

Create a relationship that links this object to an external object whose data is stored outside the Salesforce org.

Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.

Allows users to enter a date or pick a date from a popup calendar

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Email: Note that custom email addresses cannot be used for mass emails.

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'Crop' and 'New Custom Field'. It is at 'Step 3: Enter formula'. The 'Insert Field' section shows the formula: $\text{Actual_Yield_c} / \text{Total_Acre_Harvested_c}$. The 'Insert Operator' dropdown is set to '/'. The 'Functions' panel on the right shows a list of functions including ABS, ACOS, ADDMONTHS, AND, ASCII, and ASIN. The 'Quick Tips' panel on the right provides links for 'Getting Started' and 'Operators & Functions'.

Activity 3:

Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

▼ Instructions

- Legend**
- Excluded Value
- Included Value

Click button to include or exclude selected values from the dependent picklist

[illegible]

Click buttons to indicate as much as possible whether each source has a good/bad attitude

Include Values	Exclude Values
----------------	----------------

Save Cancel Preview

7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.



Search Setup

Setup Home Object Manager

profile

UsersProfiles

Didn't find what you're looking for?
Try using Global Search.

SetupProfiles

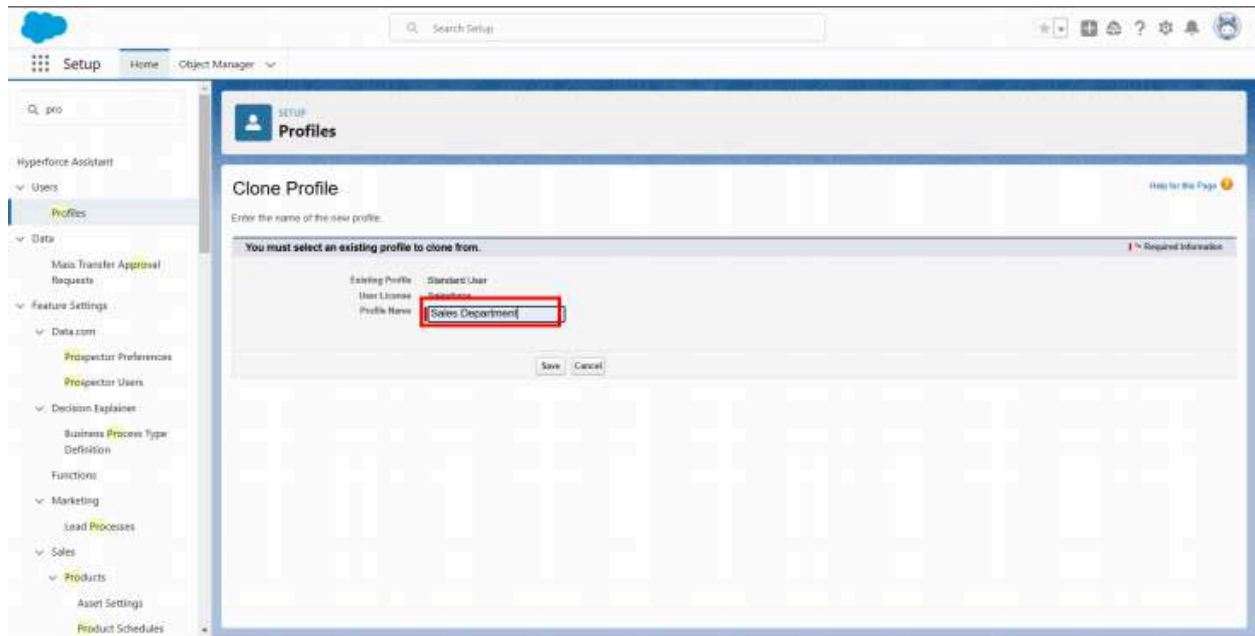
Profiles

Help for this Page

All Profiles | Add | Remove | Create New View

New Profile: 4

Action	Profile Name	User License	Custom
☐	Dev (Dev)...	Sales Development	
☐	Dev (Dev)...	Sales Manager	
☐	Dev (Dev)...	Salesforce API User System Administrator	
☐	Dev (Dev)...	System Partner User	
☐	Dev (Dev)...	System Manager	
☐	Dev (Dev)...	Standard Platform User	
☐	Dev (Dev)...	Standard Platform User	
☐	Dev (Dev)...	Standard Platform User	
☐	Dev (Dev)...	Standard User	
☐	Dev (Dev)...	System Administrator	



5. Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

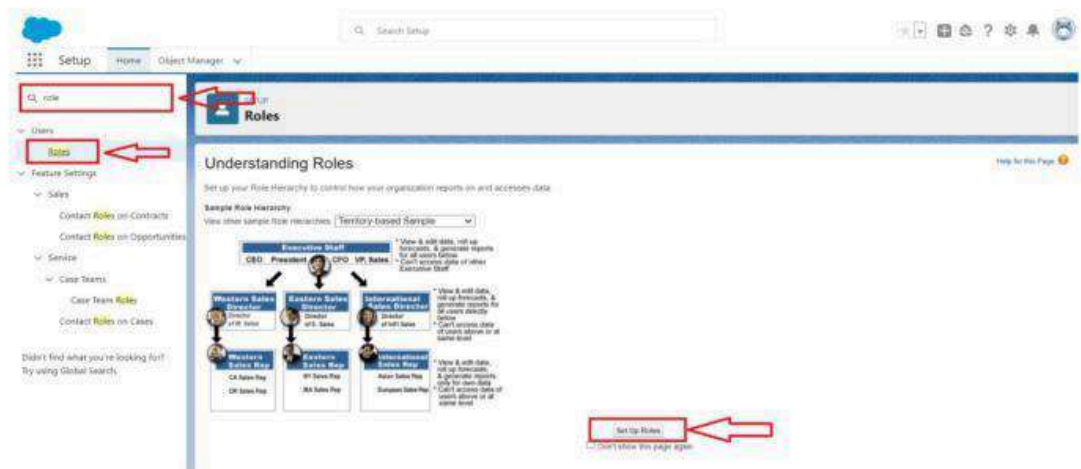
Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

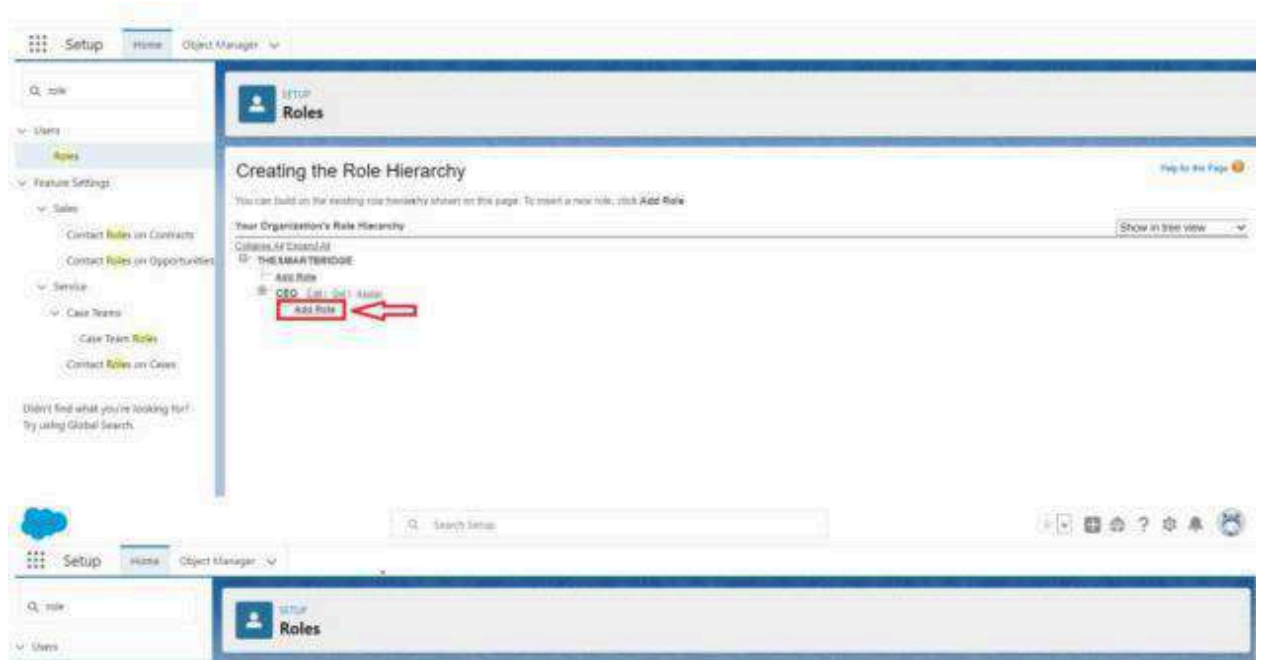
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option



3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.



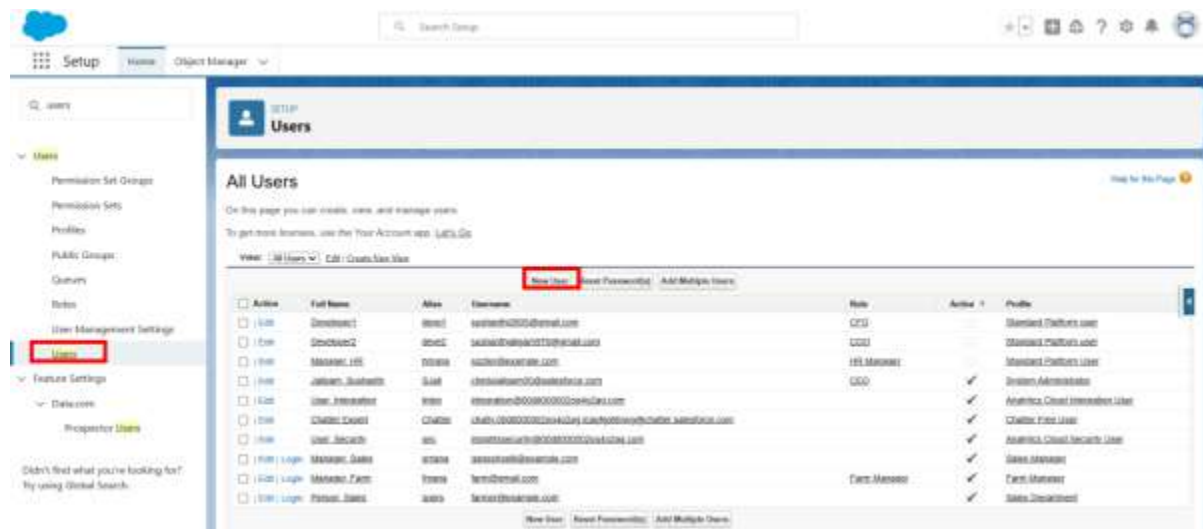
What is a user?

Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

Setup

Install

Object Manager

Search Setup

Sharing

Security

Guest User Sharing Rule Access Report

Sharing Settings

Didn't find what you're looking for? Try using Global Search.

1

Sharing Settings

Use sharing rules to make automatic decisions to your organization-wide sharing settings for defined sets of users.
Note: "Role and subordinates" includes all users in a role, and the roles below that role.
You can use sharing rules only to grant access; you can't deny access.

Label

Crop

Rule Name

Crop

Description

Step 1: Select your rule type

Criteria

Field	Operator	Value	
Crop Name	equals	100	AND
None	None	None	AND
None	None	None	AND
None	None	None	AND
None	None	None	AND

Add New Criteria

Include records owned by users who can't have an assigned role

Share with: Role From Manager

Access Level: Read/Write

Created By: System Admin 10/12/2024 2:29 pm

Modified By: System Admin 10/12/2024 2:29 pm

Save Cancel

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows the 'Edit Kesava' form in the Farm Management app. The form is titled 'Edit Kesava' and has a 'Required Information' icon. The fields are as follows:

- Farmer Name: Kesava
- Contact Number: 98262779
- State: Andhra Pradesh
- District: Eluru
- Village: Vindavada
- Age (Years): 18

The 'Save' button is highlighted with a red box. The background shows the 'Farm Management' app interface with a 'Recently Viewed' list and a 'Farmer Name' filter.

Milestone 11-Reports:

What are Reports?

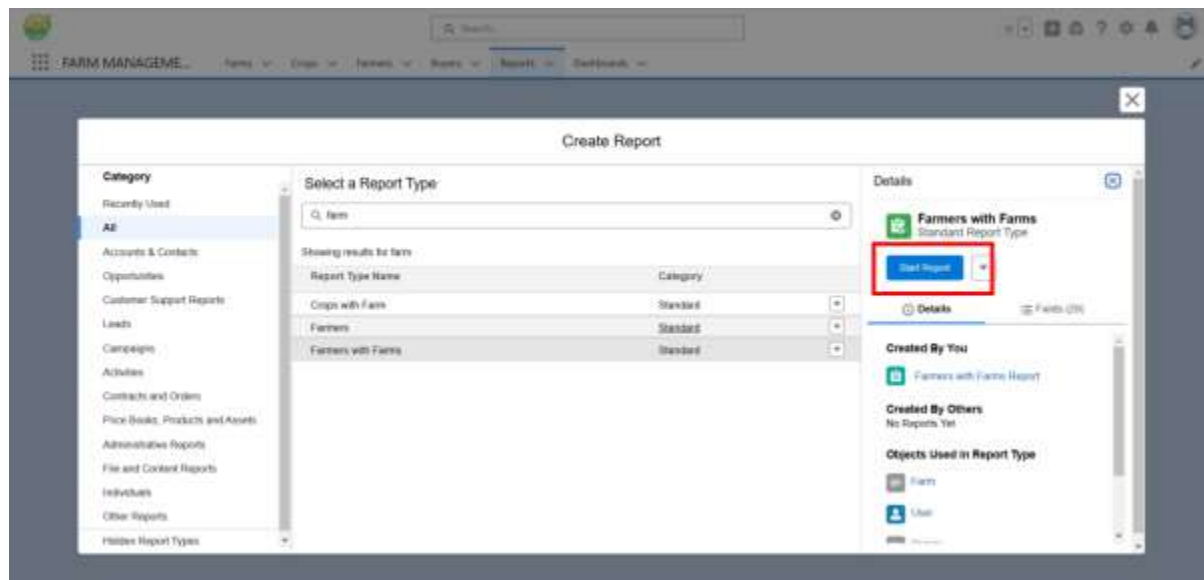
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group,Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.

The screenshot displays the 'FARM MANAGER' application interface. The top navigation bar includes 'Farms', 'Crops', 'Farmers', 'Reports', and 'Dashboards'. The 'REPORTS' section is active, showing a report titled 'Farmers with Farms Report'. The report is currently in 'Preview' mode, displaying a table of farm data.

Report Configuration:

- Groups:** Irrigation Type (Expanded)
- Columns:** Farmer: Farmer Name, Farm: Farm Name, Contact Number, Farm Size(Acres), Farm Type

Report Data Table:

Irrigation Type	Farmer: Farmer Name	Farm: Farm Name	Contact Number	Farm Size(Acres)	Farm Type
Localized Irrigation (3)	Rajesh	Paddy Farm	989430779	10	Livestock
	Rajesh	Rajesh Farm	989430779	10	Crop
	Rajesh	Wheat Farm	768202779	15	Livestock
Subtotal				25	
Total (3)				25	

The interface also includes a sidebar for 'Outline' and 'Columns', a top search bar, and a bottom status bar with options like 'Row Counts', 'Detail Rows', 'Subtotals', and 'Grand Total'.

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

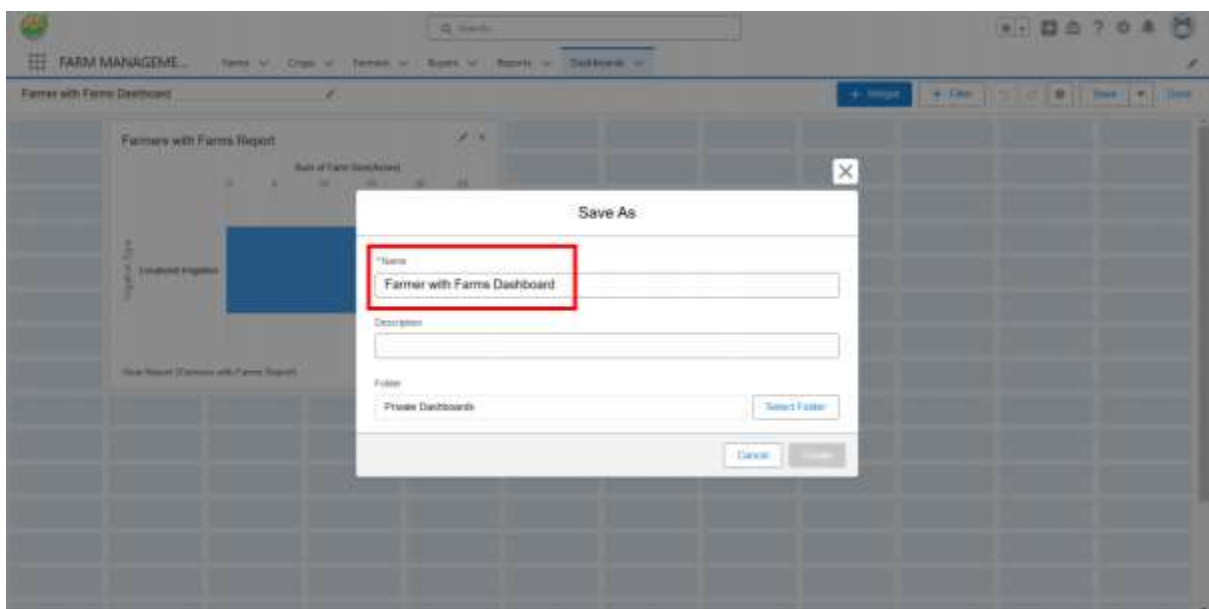
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

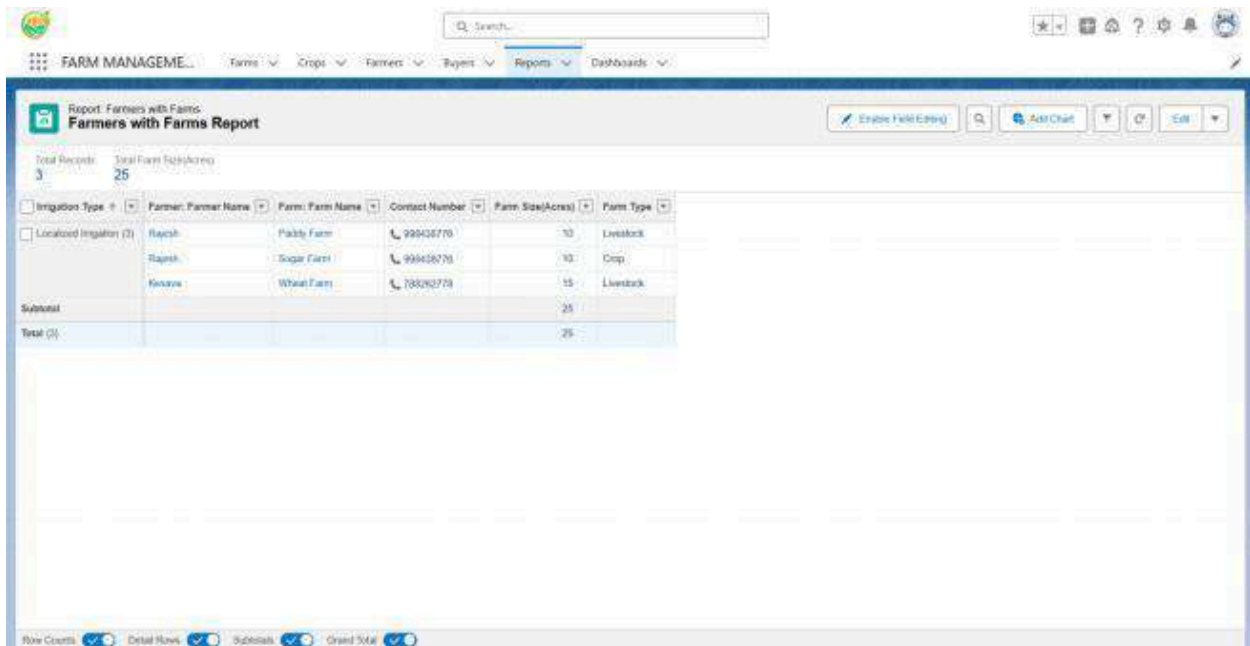
1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

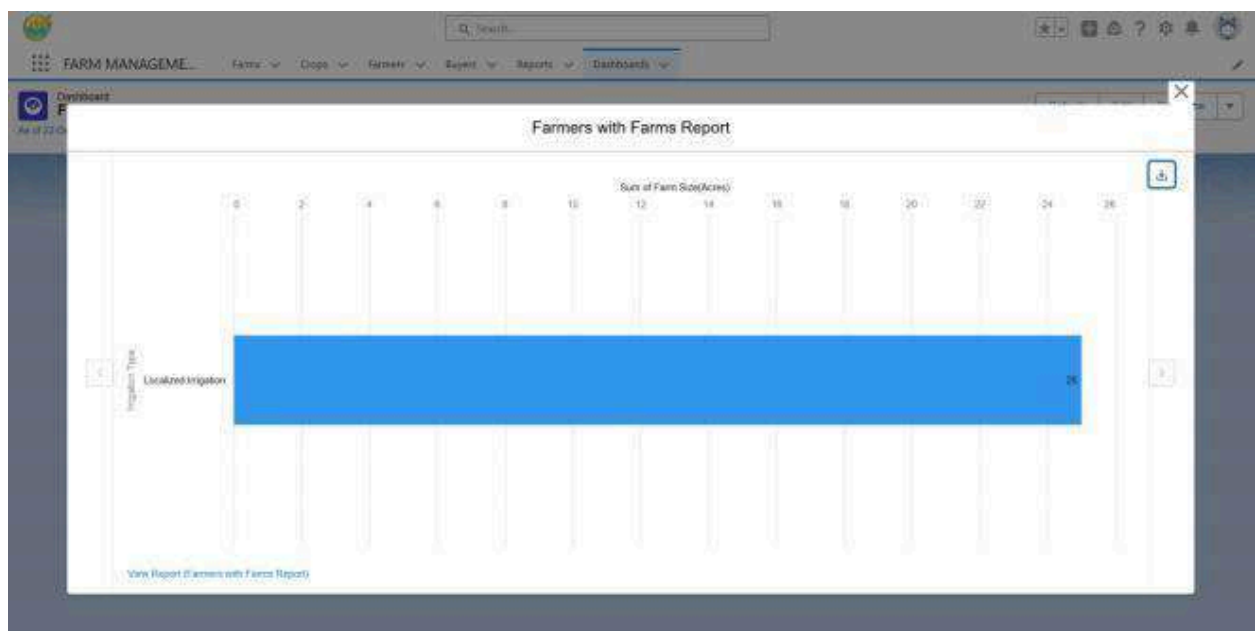


The screenshot displays the 'Farmers with Farms Report' within the Farm Management System. The interface includes a top navigation bar with tabs for Farms, Crops, Farmers, Buyers, Reports, and Dashboards. The Reports tab is active, showing a search bar and a list of reports. The 'Farmers with Farms Report' is selected, displaying a table with columns for Farmer Name, Farm Name, Contact Number, Farm Size (Acres), and Farm Type. The table shows three records: Rajesh (Paddy Farm, 10 Acres, Livestock), Rajesh (Sugar Farm, 10 Acres, Crop), and Rajesh (Wheat Farm, 15 Acres, Livestock). A summary row at the bottom indicates a total of 25 acres.

Farmer Name	Farm Name	Contact Number	Farm Size (Acres)	Farm Type
Rajesh	Paddy Farm	999436770	10	Livestock
Rajesh	Sugar Farm	999436770	10	Crop
Rajesh	Wheat Farm	785762770	15	Livestock
Subtotal			25	
Total (3)			25	

Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You

